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EDUCATION

MSc in Climate Modeling and Risk Management	Institute of Water and Flood Management (IWFM), Bangladesh University of Engineering and Technology (BUET) Dissertation: Evaluation of the 2024 Extreme Precipitation Event During Monsoon Season Over Southeastern Bangladesh Using WRF and MPAS-A Atmosphere Models. CGPA: 3.42/4.00	Sep 2024 – August 2026
BSc in Environmental Science and Resource Management	Mawlana Bhashani Science and Technology University, Tangail, Bangladesh Dissertation: Development of Emission Inventory for the Fossil Fuel-fired Power Plants in Bangladesh Using a Bottom-up Emission Inventory Approach. CGPA: 3.65/4.00	Feb 2019 - May 2024

WORK EXPERIENCE

Research Assistant Climate Change Modeling Lab Institute of Water and Flood Management (IWFM), Bangladesh University of Engineering and Technology, (BUET)	Apr 2025 - Present
Research Officer (Acting) National Oceanographic and Maritime Institute (NOAMI)	Oct 2025 - Feb 2026
Intern Envirocare	Jan 2024 - Jan 2025

PUBLICATIONS

Publications in Peer Reviewed Journals

- Chakma, A., Chakma, N., Chakma, O., Talukder, E., **Rabbany, Y.**, Jamal, S., Mahbub, H.A., and Ahmed, R. (2025) Assessing the Socio-economic Potential of Cashew Nut Farming in Rangamati, Bangladesh. *MANDALIKA: Journal of Social Science*, 3(2), 29-36. <https://journals.balaipublikasi.id/index.php/mandalika/article/view/320>

Published Abstracts in the Proceedings/Abstracts of International Conferences

- Rabbany, Y.**, Fahim, S.I., Laskor, Md. A.I.H., Dipu, S.U.A., Bhuiyan, F., and Islam, A.S. (2026) Sensitivity of heatwave simulation to radiation parameterization in WRF and MPAS-A: A case study over Bangladesh. In EGU General Assembly 2026, EGU26-16320.

MANUSCRIPTS IN PREPARATION

- Rabbany, Y.**, Fahim, S.I., and Islam, A.S. (2026) Sensitivity Analysis of WRF and MPAS-A for the 2024 Extreme Precipitation Event over Southeastern Bangladesh.
- Fahim, S.I., **Rabbany, Y.**, and Islam, A.S. (2026) Assessment of WRF and MPAS-A Performance in Simulating Lightning Activity over Bangladesh.
- Fahim, S.I., **Rabbany, Y.**, and Islam, A.S. (2026) Evaluation of Rainfall-induced Landslide over Bangladesh Using WRF-ARW Model.

RESEARCH EXPERIENCE

Research Projects

Research Assistant, "*Air Quality Management in Bangladesh*" Oct 2026 - Present
Collaborative project of BUET, Dhaka University, Department of Environment (Bangladesh), University of Washington, University at Albany, Clarkson University (USA) and IIT-Bombay (India)

Key Responsibilities: National emission inventory development, air quality monitoring and data analysis, air quality modelling, Report Writing.

Research Assistant, "*Enhancing Climate Resilience of Coastal Areas of Bangladesh with Nature-based Solutions*" Jan 2025 - Jan 2026

Funded by Global Center on Adaptation (GCA)

Key Responsibilities: Data collection and processing, Weather Modelling, Hydrodynamic Modelling, Field works, Report Writing.

Project Member, "*Carbon Emission Reduction Potential in the Thermal Power Sector in Bangladesh Using Renewable and Sustainable Energy Alternatives Towards Environmental Sustainability*" Apr 2023 - Sep 2024

Key Responsibilities: Data collection and processing, Inventory & Statistical analysis, Field works, Report Writing.

TEACHING EXPERIENCE

Short Course on Numerical Modeling in Weather and Climate Sector 03 June to 10 August 2025
Hands-on: WRF, MPAS-A Model setup and run, Visualizations of model results and optimization of physics parameters.

OPEN-SOURCE CONTRIBUTIONS

Community Contributor, WRF-MOSIT and MPAS-MAST Jan 2025 - Present
Troubleshoot model performance issues and optimize computational efficiency.
Develop and share Python scripts for meteorological data processing.
Test new model configurations and validate forecast outputs.
github.com/yeamin47/MPAS-MAST | github.com/HathewayWill/WRF-MOSIT

PROFESSIONAL SKILLS

Programming/Analysis: Python, Fortran, Bash

Numerical Weather Models: WRF-ARW, MPAS-A, WRF-Hydro

GIS and Remote Sensing: ArcGIS, ArcGIS Pro, QGIS

Machine Learning: scikit-learn, TensorFlow, PyTorch; Random Forest, XGBoost, CNN, LSTM

Modeling tools: R, MatLab, CDO, NCO

HPC: Linux, UNIX, MPI-parallel, Batch Script

Version Control: Git, GitHub

Other Software: Google Earth Engine, Origin, SPSS

ENGLISH LANGUAGE PROFICIENCY

IELTS Academic: Overall Band 7.0 (CEFR C1)

Listening 7.5 | Reading 8.0 | Writing 6.5 | Speaking 6.0

RESEARCH INTERESTS

Climate and Regional NWP Modeling || Multi-Hazard Risk Assessment and Modeling || Environmental Modeling || Extreme Weather || Machine Learning, Deep Learning and Hydroclimatic Extremes || GIS and Remote Sensing

MEMBERSHIP AND AFFILIATION

American Geophysical Union (AGU) 2026 - Present

ADDITIONAL ACADEMIC COURSES

- Joint MPAS/WRF Users Workshop 2025, organized by the Mesoscale and Microscale Meteorology Laboratory (MMM), NSF NCAR, Boulder, Colorado, USA, June 3-6, 2025.

REFERENCES

Dr. AKM Saiful Islam [MSc Supervisor]

Professor, Institute of Water and Flood Management (IWFM)

Bangladesh University of Engineering and Technology (BUET)

Email: akmsaifulislam@iwfm.buet.ac.bd

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